

Sanjana Venkatesh

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Computer Science undergraduate (expected graduation: June 2027) with backend and systems engineering experience across Python, Java, Node.js, backend APIs, web scraping/automation, and relational & cloud databases (PostgreSQL, MongoDB Atlas). Backed by two software engineering internships, 400+ DSA problems solved (LeetCode/HackerRank, CodeChef rating 1386), and independent backend/systems projects with production debugging under real deployment conditions.

EDUCATION

- **RNS Institute of Technology** Bengaluru, India
B.E. in Computer Science & Engineering *Expected Jun 2027*

EXPERIENCE

- **IEEE, KLE Section** *Jun 2025 – Oct 2025*
 - **Semantic Retrieval Backend:** Built a Node.js/Express backend matching natural-language physics questions to the correct simulation, measured by accurate selection across a ~30-simulation corpus, by generating Gemini embeddings, ranking via semantic similarity, and constraining a second Gemini call to the matched simulation's parameter schema for extraction.
 - **API Robustness & Git Workflow:** Hardened the core API against production failures, measured by structured 400/429/500 responses under a 10 req/60s rate limit, by adding input validation, prompt gating, and CORS across local/Render/Vercel origins, while reviewing PRs and leading Git/GitHub workflow for a 4-member team.
- **Samsung Innovation Campus** *Sep 2025 – Jul 2026*
 - **IoT Intern**
 - **Embedded Systems Engineering:** Built an ESP32 touchscreen productivity dashboard (SPI TFT + capacitive touch), measured by real-time LeetCode/GitHub stat syncing without UI blocking, by architecting non-blocking millisecond-timer scheduling with staggered API refresh intervals across Wi-Fi/HTTPS/GraphQL calls; selected among 800+ applicants.

PROJECTS

- **LeadLattice: AI Lead-Generation Agent (Web Scraping)** Mar 2026
Python, Web Scraping, REST APIs, Next.js, Streamlit
 - **Scraping Infrastructure & Resilience:** Built a resilient PubMed scraping pipeline in Python, measured by ~500 ranked leads per run with no unhandled request failures, by implementing pagination, rate-limiting/throttling to NCBI's req/s limit, and retry logic with exponential backoff (urllib3.Retry, 429/500–504) plus a weighted scoring algorithm across Next.js/Streamlit dashboards.
- **BleepBloop: Real-Time Chat Backend** Feb 2026
Java, Spring Boot, WebSockets, PostgreSQL, Docker
 - **Production Debugging & Real-Time Messaging:** Resolved a production-only WebSocket outage, measured by restoring real-time chat delivery over Render's HTTPS/SSL termination, by enforcing HTTPS URL resolution, adding Spring Boot CORS config, and explicitly configuring SockJS transports; also implemented per-user private messaging via STOMP user destinations and `SimpMessagingTemplate.convertAndSendToUser()`.
- **3W: Social Media Platform (Cloud-Deployed)** 2026
React, Node.js/Express.js, MongoDB Atlas, JWT
 - **Schema Design & Concurrency-Safe Interactions:** Designed a MongoDB schema for a social feed, measured by single-query feed reads (no per-post joins) with zero duplicate likes, by embedding comments within posts with compound indexes on `created_at/author_id`, and implementing idempotent like/unlike via `$addToSet/$pull` paired with a frontend request lock.

PROGRAMMING SKILLS

- **Languages:** Python, Java, JavaScript, TypeScript, C, HTML5/CSS
- **Backend & Systems:** Node.js, Express.js, Spring Boot, Django, REST APIs, Web Scraping (rate limiting, retries, parsing)
- **Databases:** PostgreSQL, MongoDB Atlas (cloud), SQLite, Relational & NoSQL Schema Design
- **Tools & Workflow:** Git/GitHub (PR workflows), Docker, Debugging, Testing, JWT, Render, Vercel
- **Problem Solving:** 400+ DSA problems solved (LeetCode, HackerRank); CodeChef rating 1386